A Smarter Way to Compare Multi-Dimentional Data Distributions

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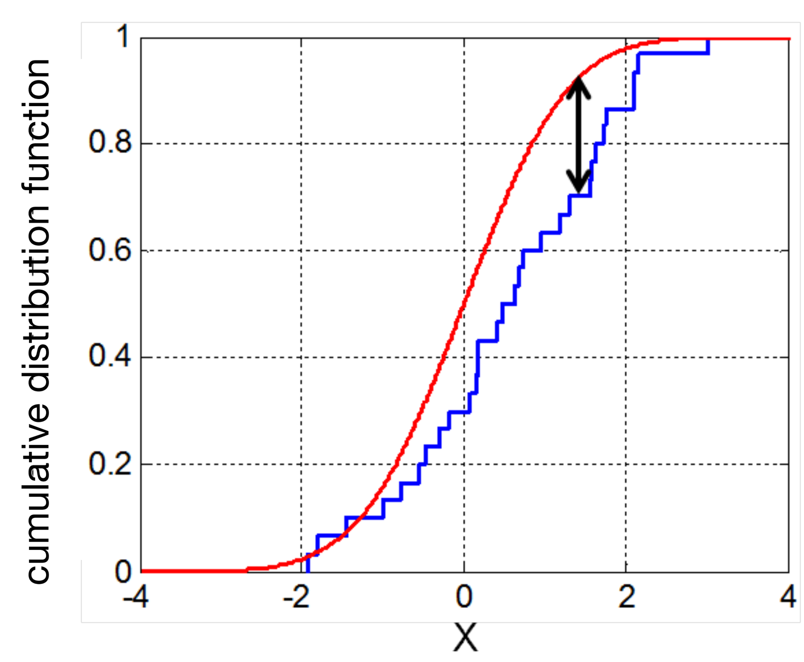
Kolmogorov–Smirnov Distance?

KS Distance is the maximum vertical distance between $f(x)$ and $g(x)$



$$D_{KS}(F,G) = \sup_x |F(x) - G(x)|$$

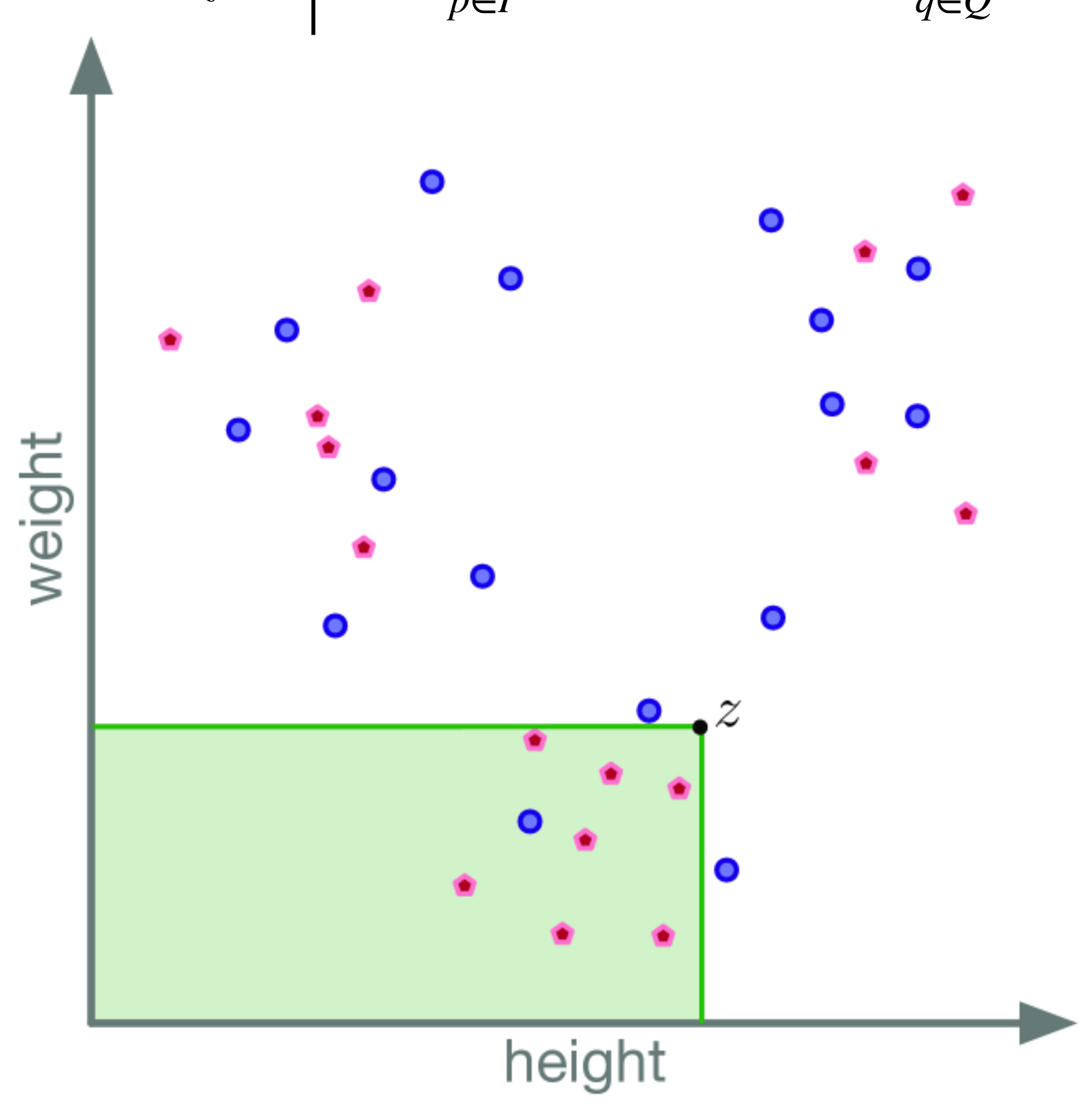
WHY THIS MATTERS

1. **Comparing distributions** is key in stats, AI, and fairness.
2. **KS test works great in 1D — but** breaks down in higher dimensions.
3. **Existing fixes (e.g., mdKS)** are untable and lack theoretical support.

OUR SOLUTION: dKS

d-dimensional Kolmogorov–Smirnov (dKS) distance formula:

$$\mathrm{dKS}(P,Q) = \sup_{z \in \mathbb{R}^d} \left| \frac{1}{|P|} \sum_{p \in P} \mathbf{1}[p \preceq z] - \frac{1}{|Q|} \sum_{q \in Q} \mathbf{1}[q \preceq z] \right|$$



ADVANTAGE OF dKS

- Comparing multi-dimensional distributions in stats, AI, ML, etc.
- **Invariant to unit** scaling (e.g., cm vs inches)
- Sample Complexity:

$$k_\varepsilon = \mathcal{O}\left(\frac{1}{\varepsilon^2}\left(d + \log\frac{1}{\delta}\right)\right)$$

Stable to sampling error

- An Integral Probability Metric (IPM)
  Satisfies **triangle inequality**
- **Efficient** to compute:
  - d = 2: as fast as in 1D
  - d = 3, 4: near-linear in sample size

ALGORITHM (in d=2)

1- Exact Algorithm:

Draw k_ε i.i.d. points:

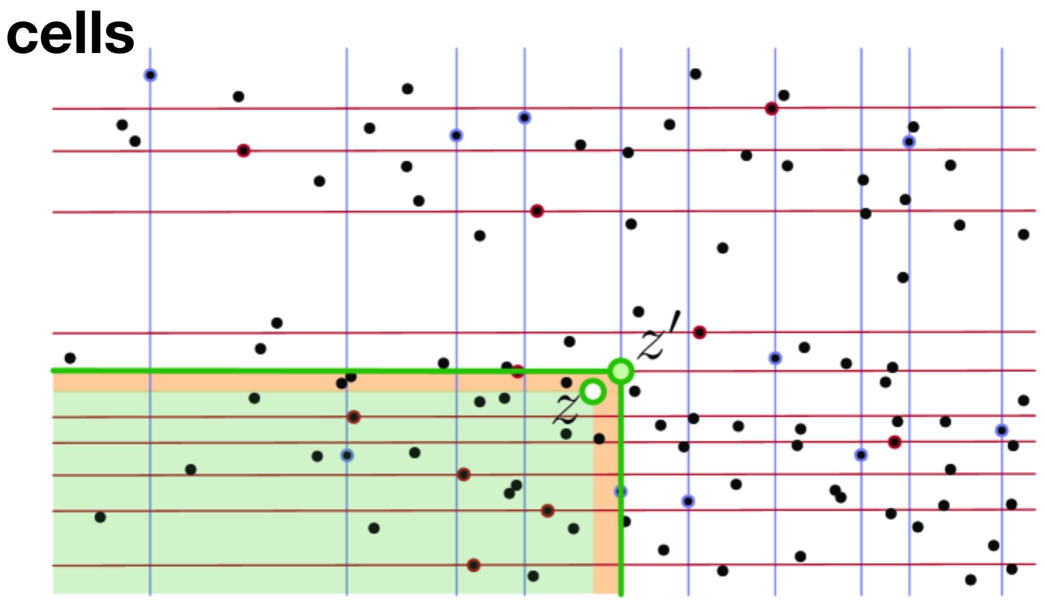
$$P \sim \mu_P, \quad Q \sim \mu_Q$$

2- Grid construction

Sort the union $P \cup Q$

Select $\sqrt{k_\varepsilon}$ evenly spaced points in each dimension

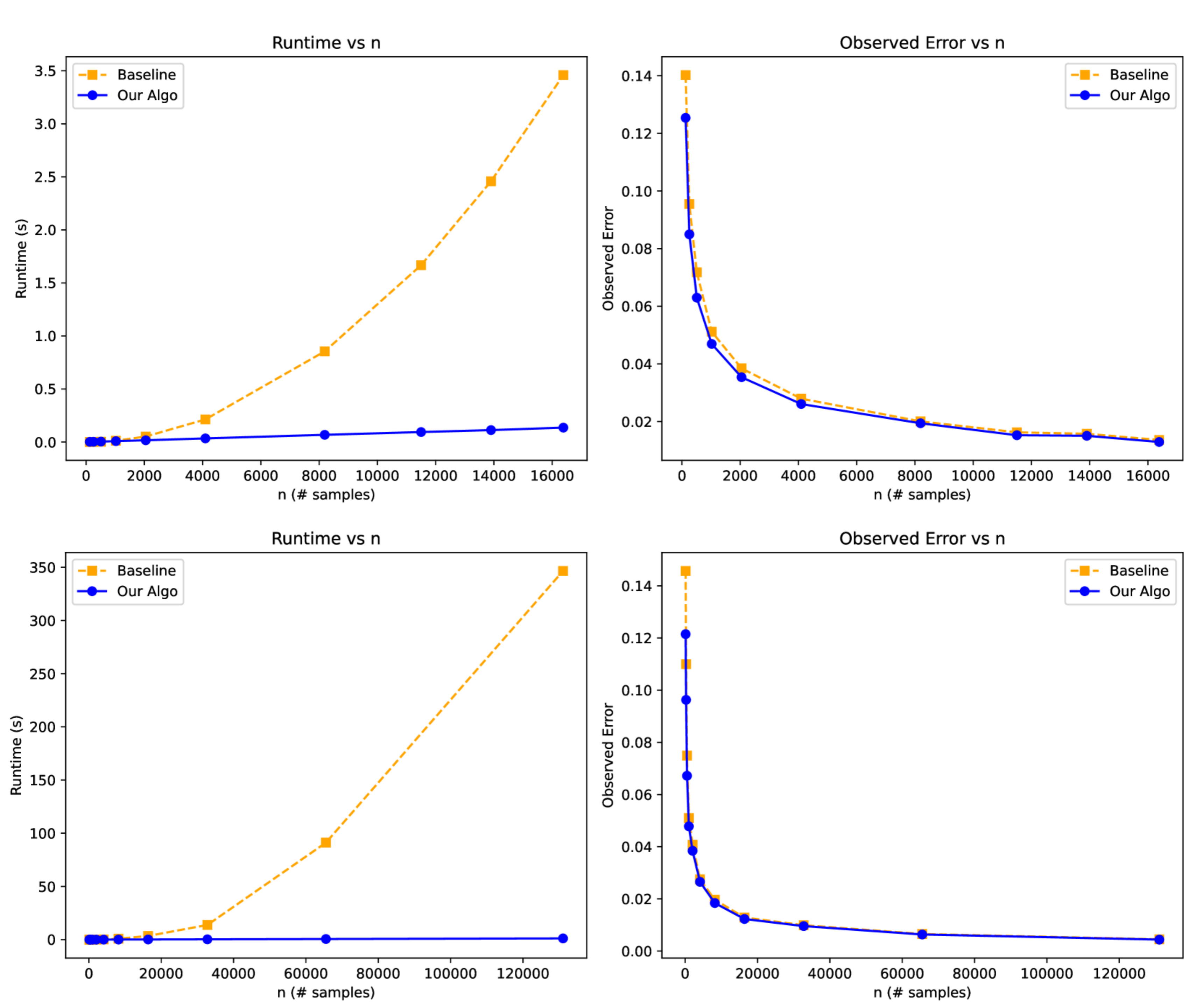
3- Round points to grid cells



4- Compute dKS

Evaluate the maximum empirical CDF difference over the induced grid rectangles

Guarantee: This induces error, on the same order as sampling noise.
But is much faster than checking all rectangles.



STATISTICAL TEST

dKS enables accurate two-sample tests with faster runtimes and stronger guarantees than mdKS or Peacock.

Citation	Test Statistic	Stabile	Precise Test	(n,δ) Test Runtime	
[25, 35]	Peacock	Yes	No	$O(n^d)$	
[10, 19]	mdKS	No	No	$O(n \log n)$	
this paper	dKS	Yes	Yes	$O(n \log n \log(1/\delta))$	$d = 2$
this paper	dKS	Yes	Yes	$O(n \log^3 n \log(1/\delta))$	$d = 3$
this paper	dKS	Yes	Yes	$O(n \log^{21+\zeta} n \log(1/\delta))$	$d = 4$